HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and developed embedded C++ software for microcontroller-based systems, implementing real-time communication protocols and device drivers for **safety-critical life safety products**.
- Implemented and debugged embedded applications, focusing on **wireless connectivity integration** for technologies including BLE and proprietary RF protocols, ensuring reliable data transmission.
- Contributed to software architecture and design reviews, providing technical input on microcontroller selection and system-level feature implementation for new product development.
- Collaborated with Hardware and Validation teams to integrate and test embedded software on new hardware revisions, performing root-cause analysis for software defects and implementing corrective actions.
- Developed and executed comprehensive unit and integration tests for embedded firmware, maintaining a high level of code quality and adherence to development standards.
- Managed software defect tracking and resolution using **Jira**, participating actively in **Agile/Scrum ceremonies** and contributing to sprint planning and technical estimations.
- Containerized embedded build environments using **Docker**, streamlining the development workflow and ensuring consistent builds across different developer machines.

SKILLS

Programming Languages – C, C++, Python, Rust

Embedded Systems – Microcontrollers, RTOS, Bare-metal, Embedded Linux, ARM Cortex-M

Connectivity & Protocols – BLE, Matter, Thread, Wi-Fi, SPI, I²C, UART, USB, RF Protocols

Development Tools – C/C++ IDEs, Debuggers, Git, Jenkins, Jira, Confluence

Methodologies – Agile, Scrum, Test-Driven Development (TDD), CI/CD

Hardware Interaction – Datasheet Interpretation, Schematics Analysis, Hardware-Software Integration

PROJECTS

Embedded Firmware for Connected Home Sensor

- Developed embedded C firmware for a new generation of smart home sensors, integrating **BLE connectivity** for seamless smartphone pairing and data reporting.
- Implemented low-power modes and optimized sensor reading routines to extend battery life, achieving an estimated **18-month operational period** on a single charge.
- Wrote unit tests and performed hardware-in-the-loop testing to validate sensor accuracy and communication reliability, ensuring compliance with product specifications.

IoT Device Fleet Management Automation (HCL Technologies)

- Created Python scripts to automate the deployment and configuration of firmware updates across a fleet of **~500 IoT devices** in a test environment.
- Developed REST APIs using **Flask** to manage device registration and status reporting, integrating with an internal device management platform.
- Integrated CI/CD pipelines using **Jenkins** to automate the build, test, and deployment process for firmware releases, reducing manual intervention by **40%**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)